

Personal Track Safety

PTS 675-000 V4 — Revision Notes

East Midlands Railway

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1. Course Aim & Objectives

This course enables you to safely negotiate the railway when:

- Lineside or On or Near the Line on Network Rail managed infrastructure
- On or Near the Line on an EMR managed Depot or Siding
- Protecting the line in the case of an emergency

Key Objectives by Section

- State when PTS applies and the difference between walking routes
- Identify what PPE is required and the difference between Lineside and On or Near the Line
- Understand track layout and key locations on the track
- Know what information is required before going on the line
- Identify hazards and the correct procedure for crossing the line
- Recognise lineside signs, refuges and telephones
- Complete an emergency call to the signaller
- Understand AC Overhead Line Equipment (OLE) and DC Conductor Rail safety
- Carry out Emergency Protection
- Identify the COSS and types of protection used when working On or Near the Line

2. Walking Routes

Walking routes are risk-assessed routes that enable personnel to move safely around Depots, Yards and Sidings.

Authorised Walking Routes (AWR)

AWRs are risk-assessed routes designated within local working instructions and may be used by personnel not Track Safety competent. They are marked with yellow lines and green pedestrian signs.

Route Colour Coding

Colour	Description / Risk Level
GREEN	Lowest risk. Within Depot Protection area, lineside, or separated from the operational railway by a permanent barrier. No specific PPE required (check LWI). Must be site inducted or conducted.
YELLOW	On or Near the Line. As a minimum: approved Hi-Vis jacket or vest + sensible footwear. Max 5 visitors per Responsible Person when On or Near the Line.
RED	High risk. Must only be used in the course of duties. Hi-Vis + authorised safety footwear mandatory. Users must be site inducted AND track safety competent (unless conducted). Max 2 visitors per Responsible Person.

Rules for All Walking Route Users

- Understand the depot layout and walking routes before using them
- Follow all local working instructions
- Walk in single file; keep to the centre of the route where possible
- Be vigilant to slipping/tripping hazards and train movements
- If two users need to pass, one must give way and move to the side
- Carry long items horizontally using a second person where possible

⚠ No objects shall be placed or moved within 2 metres of the nearest running line

3. Depot Safety

Your safety around the depot is YOUR responsibility.

General Depot Rules

- Receive and fully understand the site safety brief
- Follow all local working instructions
- Only work or walk in areas you are authorised to
- Understand the depot boundaries — know where the depot ends and mainline begins
- Obey all warning signs
- Always be aware of moving trains

Depot Protection

A loud siren and flashing strobe lights indicate that depot protection has been turned off and a train movement will commence. Move to a position of safety immediately.

Stabled Trains

⚠ Leave a MINIMUM distance of 50 feet / 15 metres when crossing between stabled trains. If not possible, take another safe route.

4. PPE, Lineside & On or Near the Line

PPE Requirements

High Visibility clothing **MUST** always be:

- Clean
- Worn correctly
- Of the approved type
- In darkness — must have reflective markings

Lineside vs. On or Near the Line

Area	Definition
Lineside	Inside the boundary fence AND more than 3 metres from the nearest running line
On or Near the Line	Inside the boundary fence AND less than 3 metres from the nearest running line
Platform	Considered On or Near the Line if engineering/technical work is within 4 feet (1.25 m) of the platform edge

Before Going On or Near the Line

- Wear correctly fitted, clean Hi-Vis clothing
- Obey all warning indications (lights, bells, notices)
- Look and listen for approaching trains
- Acknowledge the audible warning from an approaching train

⚠ You must NOT go on or near the line unless it is absolutely necessary in connection with your duties AND you have a current PTS certificate of competence.

5. Permanent Way & Track Layout

Components of the Permanent Way

Component	Description
Rail	The steel rail that trains run on
Pandrol Clip	Holds the rail to the chair or base plate. Other types include spikes and keys
Chair / Base Plate	Normally sits on a rubber pad on top of the sleeper
Sleeper	Can be made of wood, concrete or metal
Ballast	Generally granite or limestone; sits under the sleeper
Fishplate	Joins two lengths of rail together

Track Zones (Two-Track Layout)

- CESS — area outside the running rails
- FOUR FOOT — space between the two rails of one track
- SIX FOOT — space between two adjacent tracks

Track Zones (Four-Track Layout)

- CESS — outer edges
- FOUR FOOT — inside each track
- SIX FOOT — between adjacent tracks
- TEN FOOT — between the fast and slow lines

6. Track Safety Information & The 5-10 Rule

Information Required Before Going on the Line

- The speed trains are permitted to travel on each line
- The signalled direction(s) trains approach on each line
- The names of the lines

This information can be found in the SECTIONAL APPENDIX.

Train Directions

Direction	Meaning
UP Line	Towards London
DOWN Line	Away from London

Bi-Directional Lines

Some single lines are signalled to run in both directions. Extra care must be taken on bi-directional lines.

Where to Walk

If a walking route is not available, the safest place to walk is in the CESS, facing oncoming traffic.

The 5 and 10 Second Rule

⚠ You MUST be able to look up at least every 5 seconds AND reach a position of safety at least 10 seconds before the train passes.

When Walking On or Near the Line You Must:

- Look up every 5 seconds
- Watch and listen for approaching trains
- NOT allow yourself to become distracted by anything or anyone
- NOT rely on anyone else to warn you about approaching trains
- NOT rely on signals at danger to protect you

If a Train Approaches:

- MOVE to a position of safety if not already in one
- ACKNOWLEDGE any audible warning by raising ONE arm above your head
- CHECK for any other trains before moving from the position of safety

Position of Safety Distances

Line Speed	Minimum Distance from Running Line
0–100 mph	At least 4 feet / 1.25 metres
Above 101 mph	6 feet 6 inches / 2 metres

Equipment: If working with equipment and a train approaches, place equipment at least 6 feet 6 inches / 2 metres from the running line — regardless of line speed.

Walking in Groups

- Walk at least 20 yards / metres away from another person
- If 2 or more people walk together, a COSS must be appointed
- Exception: when a Designated Person (DP) is in charge of a group

7. Hazards & Crossing the Line

Common Lineside Hazards

- Approaching trains
- Uneven and slippery ballast
- Signalling cables and wires
- Oil contamination on the railhead
- Broken troughing
- Electrified third rail and overhead lines
- Rubbish
- Moving sections of points and other fittings
- Risks from toilet discharge, dead animals or hazardous chemicals

Poor Weather

- Can cause slips, trips and falls
- Can reduce sighting distances
- Can affect the safety of the line
- Wear correct PPE and report safety-of-the-line issues to the signaller immediately

⚠ **Umbrellas must NOT be used while walking Lineside or On or Near the Line.**

⚠ **Mobile phone use should be avoided. If essential, stand still in a position of safety.**

Things That Reduce Sighting Distances

- Tunnels
- Bridges
- Overhead Line Gantries
- Curves and bends
- Weather
- Foliage

Crossing the Track — Rules

- Do NOT step on the rail
- Do NOT step on the sleepers
- Do NOT cross between points
- Check for trains
- Identify a position of safety on the opposite side BEFORE crossing
- Cross directly in a straight line — NOT diagonally

Points and Crossings

Extra care is needed at points and crossings. Do not cross between the points. Where possible find an alternative route.

Caught Between Two Adjacent Lines

⚠ If trains are approaching on BOTH adjacent lines and you cannot reach a position of safety: LAY DOWN IN THE 6 FOOT or 10 FOOT. NEVER lay down in the 4 foot!

Entering Tunnels

- Only enter a tunnel if it is safe to do so
- Ensure you have a torch or lamp
- Identify refuge points before entering
- If a train approaches and you cannot reach a refuge — lay down in the 6 foot, 10 foot or cess. NEVER the 4 foot.

8. Lineside Signs, Refuges & Telephones

Sign Colour Meanings

Colour	Meaning
RED	Prohibition — something you are forbidden from doing
BLUE	Mandatory — something you must do
YELLOW	Danger warning
GREEN	Information

Key Signs

- Prohibition Sign (red) — Do not pass beyond this point while trains are running. A position of safety cannot be reached safely.
- Limited Clearance — No position of safety on this side for the length of the structure. Do not enter when trains are running.
- No Refuge — No position of safety on this side, but there are refuges on the opposite side.

Refuges

A refuge is an area that is safe to stand when a train passes. They can be found:

- Built into a cutting
- In a tunnel
- Over an embankment
- On a viaduct / parapet

Mile Markers & Structure Identification

- Mile Markers are placed at $\frac{1}{4}$ mile (440 yard) intervals
- 1 chain = 22 yards; 1 mile = 80 chains
- Every structure has a structure plate with a number — use these to identify your location in an emergency

Types of Telephone

Telephone Type	Location / Use
Signal Post Telephone (SPT)	Found at signals. Picking up connects directly to the signaller.
Lineside Telephone	Direct communication with the signaller. Usually found on

	station platforms.
Level Crossing Phone	Found at level crossings. Connects directly to the signaller.
Help Points	On platforms. GREEN button = Emergency services. BLUE button = National Rail Enquiries.
Limited Clearance Telephone	Must NOT be used except in an emergency or when advised the adjacent line is blocked.

9. Emergency Calls & Communication

Communication Protocol — ABC

Always use the ABC protocol when making calls relating to safety. Use the Phonetic Alphabet when conveying messages about safety.

Example: Signal identity CE105 = "Charlie Echo — One Zero Five" (nought is read as ZERO).

When to Make an Emergency Call

- Train derailment or accident
- Large animals on the line
- Track defect
- Pooling or flowing water that might affect structures or earthworks
- Damage to railway structures (bridges, tunnels, OLE)
- Lineside fires

Standard Emergency Call — What to State

- "This is an emergency call"
- Check who you are speaking to
- Who you are — job title and company
- Where you are calling from
- What has happened
- Exact location of the incident
- Which line(s) are obstructed
- Whether emergency services are required
- Ask for your message to be REPEATED back
- Stay on the line and wait for further instructions

Dangerous Goods Emergency Call

⚠ Start your call with: "This is a rail dangerous goods emergency." State the United Nations Number from the Hazard Warning plate on the vehicle.

Example Emergency Call

"This is an emergency call. Am I speaking to the signaller? My name is Joe Smith, I am a senior conductor for East Midlands Railway. I am calling you from Signal Post Telephone — Lima Romeo One Two Three. A train has derailed [headcode if known] on the Up Main, nearest Milepost is Ninety Nine and a quarter. I require all lines to be blocked. I require all emergency services. Please repeat back."

Danger to Trains — What to Report

If you see something unsafe or dangerous on a passing train, report to the signaller immediately. Examples:

- A door not closed
- An insecure load
- A vehicle on fire
- A hot axle box
- Unlit or missing tail lamp / headlamp
- Hazard warning or taillights displayed on the front of a train
- Train sounding the distress signal (continuous series of long blasts on the horn)

10. AC Overhead Line Equipment (OLE)

Key Components

- Catenary — the top supporting wire
- Dropper — connects catenary to contact wire
- Contact Wire — the wire the pantograph runs along
- Along Track Conductor — runs the length of the route

Voltage

⚠ OLE and all its attachments are electrified at 25,000 VOLTS AC. Always treat as LIVE.

Safe Distance from OLE

⚠ Keep at least 2.75 metres (9 feet) from the OLE and its equipment at all times.

OLE Structure Identification

- Gantry types: Cantilever, Portal, Head Span
- Each gantry has a unique ID plate. Old style: letters = route, first numbers = mileage, second numbers = gantry number.
- New style (black text): ELR + route km + metres + location relative to track (e.g. UF = Up Fast)

Structure Bonds

- Each OLE structure is connected to the running rail via bonds
- Red Bonds return current to earth — must ALWAYS be considered as live
- Report defective bonds stating the colour

Cant Rail & Climbing

- The Cant Rail is the orange stripe above the windows running the length of the train
- Persons and equipment must not go above the Cant Rail

⚠ Never climb above cab floor level!

OLE Safety Rules

- Always consider OLE and its attachments to be live and extremely dangerous
- Do not touch or come near live OLE
- Do not allow clothing or objects to touch or come near live OLE, pantographs or roof-mounted equipment
- Only ladders made of wood or other non-conducting material may be used near OLE

- Long articles must be carried horizontally and must not come within 9 feet (2.75 m) of OLE

Electricity Company Power Lines on the Track

⚠ Do not get closer than 5 metres / 5 yards / 15 feet. These can be electrified at 400,000 V. Make an emergency call to the signaller immediately and get the lines blocked.

OLE Electrical Emergencies — Contact the ECO

Contact the Electrical Control Operator (ECO) directly or via the Signaller and request an emergency switch off if:

- A derailment
- A lineside fire or fire on a rail vehicle
- A person in contact with OLE
- Damaged OLE
- Any other accident or emergency

OLE Emergency Call to ECO — What to State

- "This is an electrical emergency call"
- Check who you are speaking to
- Who you are — job title and company
- Where you are calling from
- Details of the lines concerned
- The number on the nearest overhead line structure
- The reason for requesting emergency switch off
- Whether a person is in danger from OLE
- Whether emergency services are awaiting
- Ask for message to be repeated back; stay on the line

PICEE

If the ECO agrees to the switch off, they will designate a PERSON IN CHARGE OF THE ELECTRICAL EMERGENCY (PICEE).

Rescuing a Person Near OLE

The OLE must be switched off and earthed before you attempt rescue if the person is within 2.75 m (9 feet), unless ALL of the following are true:

- The person is alive and can be reached from ground level
- An emergency switch off is in place
- The person is more than 600 mm (2 feet) from the OLE and anything touching it
- You and anything you are holding will not come within 600 mm (2 feet) of OLE

11. DC Conductor Rail (3rd Rail)

Overview

The Conductor Rail (3rd Rail) is an additional rail placed on sleeper ends outside the running rails. Some lines shared with London Underground have an additional 4th rail for current return.

Voltage

⚠ The Conductor Rail is energised at 650–750 V DC. Treat as LIVE at all times.

Safe Distance from Conductor Rail

⚠ You and anything you are holding must be at least 1.5 metres / 5 feet from live Conductor Rail Equipment (CRE) or associated equipment.

Key Features

- Sits on an insulator pot — sits higher than the running rails
- Conductor Rail Gaps split sections of the DC system
- Collection Devices (collector / contact shoes) on trains make contact with the CRE — must be considered LIVE even when not in contact

DC Safety Rules — You Must NOT:

- Touch or step on CRE
- Allow clothing, tools or objects to touch CRE
- Step between the conductor rail and the adjacent running rail
- Touch broken or displaced CRE
- Touch the collector shoe on any train
- Step into flood water which may be in contact with CRE
- Direct a jet of water or liquid onto CRE

Best Places to Cross a Line with CRE

- Where there is a Gap in the conductor rail
 - On a bridge
 - Designated walkway
 - Where there are guard boards
 - Where there is a cover placed over the rail
- ✓ If none of the above is available, step over both rails in one movement.

DC Electrical Emergency — Contact the ECO

Request an emergency switch off if you become aware of:

- A derailment or lineside fire
- A fire on a rail vehicle
- A person in contact with CRE
- Damaged CRE
- An accident or emergency passenger evacuation from a train

⚠ Remember: trains can still move even after an emergency switch off. Always contact the signaller to arrange a line block for anything affecting safety of the line.

Rescuing a Person from CRE

Ensure CRE is switched off before attempting rescue of a person touching or within 300 mm (1 foot) of CRE. Only attempt rescue without switch off if ALL of the following are true:

- The person is alive
- You cannot get the CRE switched off at the time
- You can cover your hands with something dry and non-conducting
- You can stand on dry, non-conducting material
- You are not using metal objects

12. Emergency Protection

What is Emergency Protection?

If the line becomes unsafe, carry out Emergency Protection if you have the equipment available. Every train cab carries:

- 10 Detonators
- 2 Red Flags
- 2 Track Circuit Operating Clips

Detonators

- Small explosive charges placed on the railhead
- When a train wheel passes over them, they produce an audible explosion to warn the driver
- Box of 10; life span of 5 years — handle with care
- Place on TOP of the railhead, 20 metres apart
- Fasten clips under the upper flange of the railhead

⚠ Stay at least 30 metres (30 yards) away from detonators if you expect them to explode.

- In a 3rd rail area — always place on the rail FURTHEST from the conductor rail

Track Circuit Operating Clips

- Placed correctly on the running line to operate the track circuit and maintain signals at DANGER
- ONE TIME USE ONLY

⚠ MUST ALWAYS be used when carrying out Emergency Protection — even on lines NOT fitted with track circuits, and even on lines fitted with Axle Counters.

- In DC 3rd rail area: place FIRST clip on the line furthest from conductor rail first; remove the clip CLOSEST to conductor rail first

Axle Counters

Axle counters detect clear/occupied track sections using wheel sensors. They are being fitted in place of track circuits when lines are upgraded. Track Circuit Clips must still always be used for Emergency Protection on these lines.

Hand Danger Signals

Condition	Signal
Day	Two arms raised above your head OR red flag waved
Night	Red lamp OR any coloured light waved vigorously

Emergency Protection Procedure — Single Line

- Place Track Circuit Clips on all obstructed lines (and on your line if the train has derailed)
- Walk 2 km (1¼ miles) from the obstruction
- Place 3 detonators, 20 metres apart

Emergency Protection — Multi-Track Line

- Place Track Circuit Clips on each obstructed line
- Walk 2 km from the obstruction
- Place 3 detonators, 20 metres apart
- Walk 2 km from the obstruction on every other line and place 3 detonators, 20 metres apart

If a Train Approaches Before Full Protection Distance is Reached

- Immediately place down 3 detonators
- Show a hand danger signal to the approaching train from a position of safety

If You Reach an SPT or Signal Box Before Full Distance

- Place 3 detonators in front of the SPT or signal box
- Speak to the signaller

✓ You do NOT need to continue to the full distance if the signaller confirms full signal protection is being provided.

Tunnel Entrance

- Place 3 detonators at the tunnel entrance
- If the full protection distance falls within the tunnel, proceed through to the far end and place 3 detonators there

⚠ **NEVER place detonators inside a tunnel.**

Diverging Junction

- Place 3 detonators before reaching the junction
- Decide the order in which to protect each line
- Continue for the remaining 1¼ miles or until you contact the signaller
- If no contact made, return to the junction and protect the other line

13. Working On or Near the Line — COSS

Controller of Site Safety (COSS)

One COSS is appointed to take charge of a group (2 or more people) when work is carried out on the railway.

COSS Responsibilities:

- Plan and put in place a safe system of work
- Take overall responsibility of the workforce and task
- Assess risks and put in control measures
- Deliver the work safely
- Hand back after completion

Types of Protection Used by a COSS

Type	Description
Safeguarded	All lines blocked — no train movements, OR movements only at extreme caution (max 5 mph)
Fenced	Physical barriers act as a visual aid to ensure correct positions of safety are maintained. Workforce must stay on the safe side.
Separated	Work site is at least 6 ft 6 in / 2 metres from the nearest open line. A site warden is appointed by the COSS to keep the workforce within the safe zone.
ATWS	Automatic Track Warning System — automated warning of approaching trains
LOWS	Lookout Operated Warning System — lookout triggers the warning system
Lookout	A person appointed by the COSS to warn the workforce of an approaching train, who then move to a position of safety. Higher risk method; restrictions apply.

Quick Reference — Key Numbers

Fact	Value
Lineside definition	More than 3 metres from the nearest running line (inside boundary)
On or Near the Line definition	Less than 3 metres from the nearest running line (inside boundary)
Platform On or Near the Line	Within 4 feet / 1.25 m of platform edge (for engineering work)
Position of safety (0-100 mph)	At least 4 feet / 1.25 metres from running line
Position of safety (101+ mph)	At least 6 ft 6 in / 2 metres from running line
Equipment position of safety	At least 6 ft 6 in / 2 metres from running line (all speeds)
5-10 Rule	Look up every 5 seconds; reach safety 10 seconds before train passes
Walking in groups	At least 20 yards/metres apart; COSS needed for 2+ people
Stabled train crossing gap	Minimum 50 feet / 15 metres
OLE safe distance	2.75 metres / 9 feet
OLE voltage	25,000 V AC
OLE rescue — safe distance	More than 600 mm (2 feet) from OLE
Conductor Rail voltage	650–750 V DC
Conductor Rail safe distance	1.5 metres / 5 feet
Conductor Rail rescue distance	300 mm (1 foot)
Walking route max visitors (Yellow)	5 visitors per Responsible Person
Walking route max visitors (Red)	2 visitors per Responsible Person
Emergency protection distance	2 km / 1¼ miles from obstruction
Detonator spacing	3 detonators, 20 metres apart
Detonator safe distance	At least 30 metres away if expecting explosion
No objects near line	No objects placed within 2 metres of nearest running line
Electricity company power lines	Do not get closer than 5 metres / 5 yards / 15 feet (up to 400,000 V)

Cant Rail	Orange stripe above windows — do not go above it; never climb above cab floor level
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